

Rishi Raj

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Education

Trident Academy of Technology (TAT), Bhubaneswar

2023 – 2027

B.Tech in Computer Science and Engineering (Data Science)

CGPA: 7.48/10 (avg. of 6 sem.) | 7th Semester

Technical Skills

Languages: Python, SQL

Data Analysis & Statistics: Data Cleaning, Exploratory Data Analysis (EDA), Statistics, RFM Analysis, Customer Segmentation, Time Series Analysis, Data Visualization

Databases & BI Tools: PostgreSQL, Microsoft Power BI, Microsoft Excel, Matplotlib, Seaborn

ML Libraries: Pandas, NumPy, Scikit-learn, XGBoost, LightGBM

Tools & Deployment: Jupyter Notebook, Git, GitHub, Flask, Streamlit, Render

Projects

E-Commerce Customer Retention & CLV Analytics | PostgreSQL, Python, Pandas, Power BI

GitHub

- Designed a PostgreSQL schema with 8+ relational tables to load and query ~100K orders (2016–2018) from the Olist Brazilian E-Commerce dataset, verifying row counts and data integrity.
- Performed data cleaning and exploratory analysis in Python/Jupyter, then ran customer-level RFM (Recency, Frequency, Monetary) analysis to build threshold-based segments (e.g., At Risk, Loyal/Repeat Customers, Lost/Churned).
- Surfaced a ~3% repeat-purchase rate as a key retention insight; built a 2-page interactive Power BI dashboard with KPI tracking, revenue trends, and state-level cross-filtering.

E-Commerce Review Intelligence Dashboard | Python, Selenium, PostgreSQL, HF Transformers, Streamlit, Supabase

GitHub | Demo

- Built an end-to-end pipeline scraping 977 Flipkart smartwatch reviews across 5 competing brands (Selenium + BeautifulSoup), stored in a relational PostgreSQL schema (Supabase) linking raw and processed review data.
- Applied a pretrained DistilBERT sentiment model to flag rating-sentiment mismatches, surfacing a ~2x higher mismatch rate for one brand invisible from star ratings alone.
- Delivered an interactive Streamlit dashboard with brand/product filters, comparison visualizations, and keyword search; deployed with environment-variable-based credentials and an automated Supabase keep-alive workflow via GitHub Actions.

Retail Sales Forecasting (Time Series) | Python, Pandas, SARIMA, XGBoost, LightGBM, Flask

GitHub | Demo

- Analyzed ~1.02M daily sales records across 1,115 Rossmann stores (~2.5 years, 2013–2015), merging store-level attributes (type, assortment, competition, promotions, holidays) with sales data.
- Engineered lag and rolling-window features with leakage-safe validation (TimeSeriesSplit) to compare SARIMA against XGBoost and LightGBM forecasting models.
- Delivered ~9.49% MAPE with XGBoost, a ~34% MAE improvement over the SARIMA baseline (same-store comparison), and built an interactive Flask forecasting application.

Credit Risk & Loan Default Prediction | Python, Pandas, Scikit-learn, XGBoost, Flask, Render

GitHub | Demo

- Cleaned and prepared ~150K borrower records (Kaggle “Give Me Some Credit”), handling missing values, capping outliers, and removing sentinel/placeholder codes to ensure data quality.
- Engineered features (TotalTimesLate, IncomePerDependent) and evaluated Logistic Regression, Random Forest, and XGBoost, handling class imbalance and tuning classification thresholds.
- Reached ~0.86 ROC-AUC and ~77% recall on defaulters; deployed the scoring model as a Flask REST API on Render with a live demo.

Hospital Readmission Prediction | Python, Pandas, Scikit-learn, XGBoost, Flask, Render

GitHub | Demo

- Cleaned and analyzed ~101K patient records (UCI Diabetes 130-US Hospitals dataset, ~50 features) to study drivers of hospital readmission.
- Engineered diagnosis-category and prior-visit features, then compared Logistic Regression, Random Forest, and XGBoost, identifying `number_inpatient` as the leading risk factor.
- Achieved ~0.684 test ROC-AUC; deployed the model via a Flask REST API on Render and documented data ethics/privacy considerations in the project README.

Multi-Modal Transit & Logistics Route Optimization | Python, NetworkX, OR-Tools, OSMnx, Folium, Streamlit, Render

GitHub | Demo

- Extracted a real 6km-radius road network for Indore (~30K nodes, ~81K edges) via OSMnx, and solved a Capacitated Vehicle Routing Problem (CVRP) for a 4-vehicle fleet across 40 customer locations using Google OR-Tools.
- Benchmarked OR-Tools against a Nearest-Neighbor heuristic (21.8% shorter total route distance) and Dijkstra vs. A* on shortest-path queries (A* ~3.25x faster), quantifying the distance-vs-fleet-utilization trade-off from capacity constraints.
- Visualized optimized routes on an interactive Folium map and deployed a live Streamlit dashboard on Render with per-vehicle filters and route metrics.

Leadership Experience

Speech Weavers Club, Trident Academy of Technology

Jul 2024 – May 2025

Core Member

- Managed the club’s social media presence across LinkedIn, Instagram, and Facebook, creating and publishing posts covering club activities, debates, and events for an audience of 500+ college students.
- Worked within the club’s core team, coordinating on event promotion and regular audience communication alongside members handling other responsibilities.

Certifications

Google (Coursera), Mar 2026: Foundations of Data Science; The Power of Statistics; Regression Analysis: Simplify Complex Data Relationships; The Nuts and Bolts of Machine Learning

Knowledge Accelerators (Coursera), May 2026: Excel Basics for Beginners: Workbooks, Formatting, Charts and Core Skills